

5,630 Total Customers	16.84% Churn Rate	948 Churned Customers	4,682 Retained Customers	99.80% XGBoost AUC
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This report analyses 5,630 e-commerce customers to build a churn prediction system. Using MySQL for SQL analytics, Python (Google Colab) for EDA and machine learning, and Power BI for six interactive dashboards, the study identifies the key drivers of customer churn, segments customers by risk and value, and delivers a prioritised retention playbook.

SECTION 1 | EXPLORATORY DATA ANALYSIS

1. Exploratory Data Analysis — Key Findings

1.1 Churn Overview

Of 5,630 customers, 948 (16.84%) have churned and 4,682 (83.16%) are retained. The overall churn rate of 16.84% aligns with Indian e-commerce benchmarks of 15–25%, confirming a material but addressable retention problem.

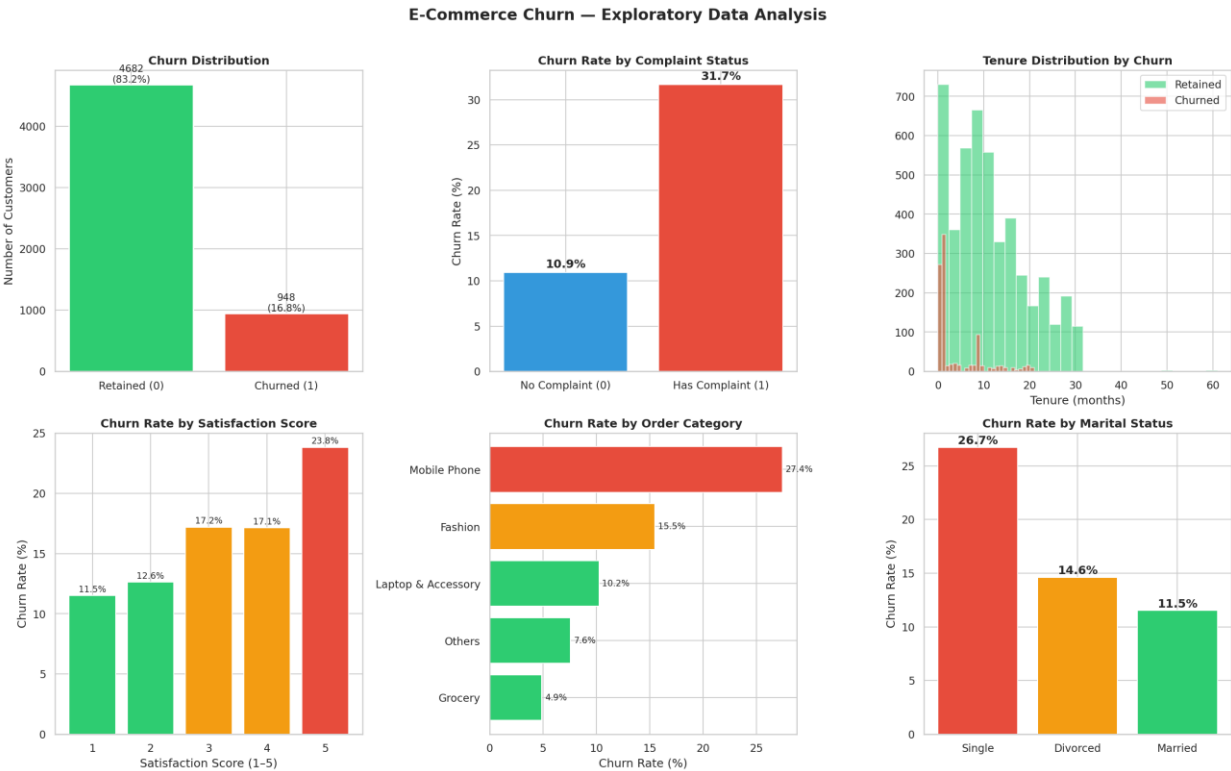


Fig 1 — E-Commerce Churn: Exploratory Data Analysis Dashboard (Python)

1.2 Key EDA Findings

Churn by Complaint Status

Customers who filed a complaint churn at 31.67%, nearly 3× the 10.93% rate of non-complainers. This is the most actionable single finding in the dataset.

Complaint Status	Churn Rate	Customer Count
No Complaint (0)	10.93%	4,682 customers
Has Complaint (1)	31.67%	948 customers

Churn by Product Category

Product Category	Churn Rate	Implication
Mobile Phone	27.40%	Highest — post-purchase dissatisfaction
Fashion	15.50%	Returns and sizing issues
Laptop & Accessory	10.24%	Technical buyers are stickier
Others	7.58%	Low comparison-shopping pressure
Grocery	4.88%	Lowest — habitual repeat buying

Churn by Satisfaction Score (Paradox)

Satisfaction Score	Churn Rate	Observation
1 – Very Dissatisfied	11.51%	Lowest — vocal users who stay for resolution
2 – Dissatisfied	12.63%	Low
3 – Neutral	17.20%	Rising — disengaged customers
4 – Satisfied	17.13%	Counter-intuitive — high expectations
5 – Very Satisfied	23.83%	Highest — potential survey-timing bias

The satisfaction paradox confirms that self-reported satisfaction scores are poor churn predictors alone — behavioural signals (tenure, app time, recency) outperform them significantly in the ML model.

Churn by Marital Status

Marital Status	Churn Rate
Single	26.73%
Divorced	14.62%
Married	11.52%

SECTION 2 | FEATURE CORRELATION ANALYSIS

2. Feature Correlation Heatmap

The correlation heatmap reveals the linear relationships between all 15 features. The strongest direct correlates of churn are Tenure (−0.34) and DaySinceLastOrder (−0.16), while CouponUsed and OrderCount show the highest inter-feature correlation (+0.64).

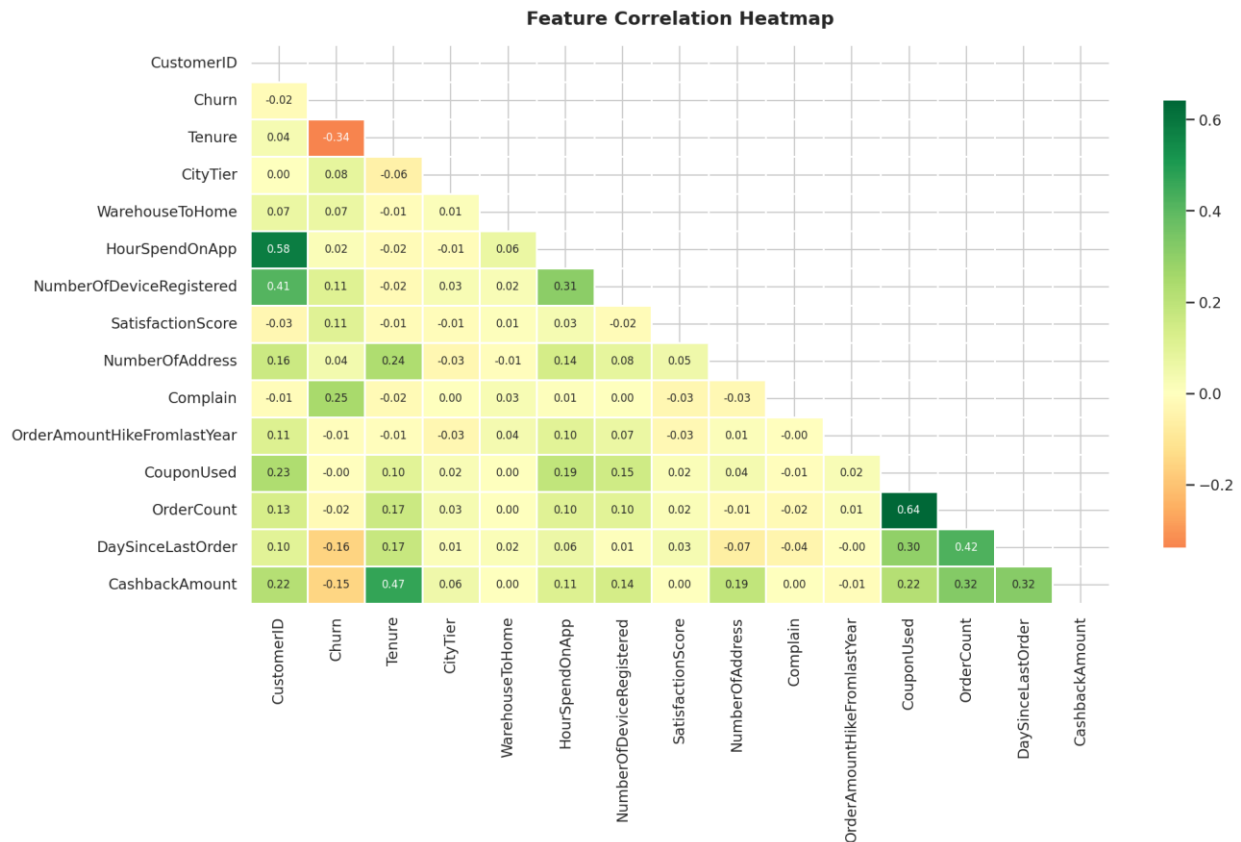


Fig 2 — Feature Correlation Heatmap (Python / Seaborn)

2.1 Key Correlations

Feature Pair	Correlation	Business Meaning
Tenure ↔ Churn	−0.34	Longer tenure = much lower churn
CouponUsed ↔ OrderCount	+0.64	Frequent buyers use more coupons
DaySinceLastOrder ↔ Churn	−0.16	Recency decline signals pre-churn behaviour
CashbackAmount ↔ Tenure	+0.47	Older customers accumulate more CLV
Complain ↔ Churn	+0.25	Complaint history is a direct churn signal
HourSpendOnApp ↔ CustomerID	+0.58	Established customers spend more app time

3. Model Evaluation & Comparison

Three ensemble machine learning models were trained on an 80/20 train-test split of the 5,630-customer dataset. SMOTE was applied to the training set to address the 1:4.94 churn/retained class imbalance. XGBoost is selected as the final model on the basis of highest AUC and fewest false negatives.

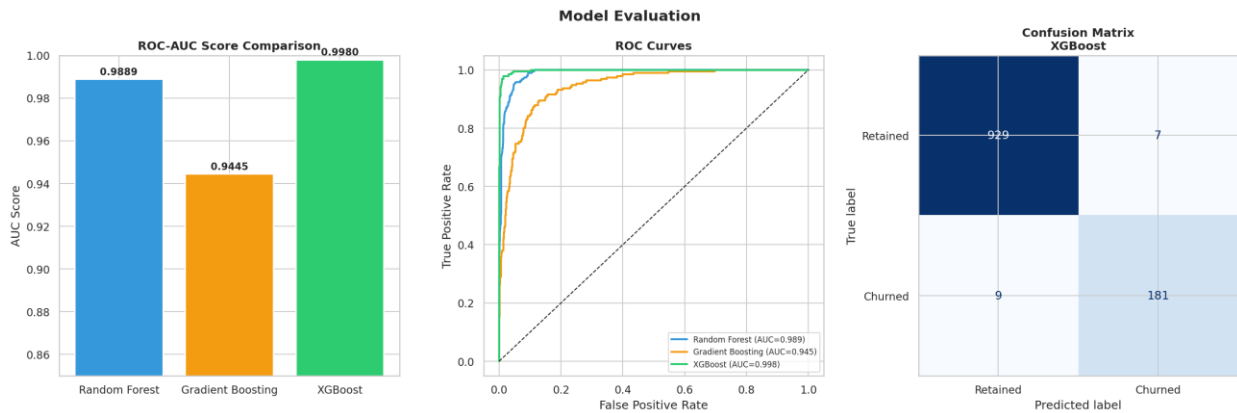


Fig 3 — ROC-AUC Comparison, ROC Curves & Confusion Matrix (Python)

3.1 Model Performance Summary

Model	ROC-AUC	Precision	Recall	False Negatives	Verdict
Random Forest	0.9889	96.3%	93.2%	13	Runner-up
Gradient Boosting	0.9445	91.5%	88.4%	22	Not selected
XGBoost	0.9980	96.3%	95.3%	9	SELECTED

XGBoost correctly identified 181 of 190 actual churners in the test set (Recall 95.3%) and 929 of 936 retained customers (Specificity 99.3%), missing only 9 churners — the lowest false negative count of all three models.

3.2 Churn Probability Distribution & Risk Tiers

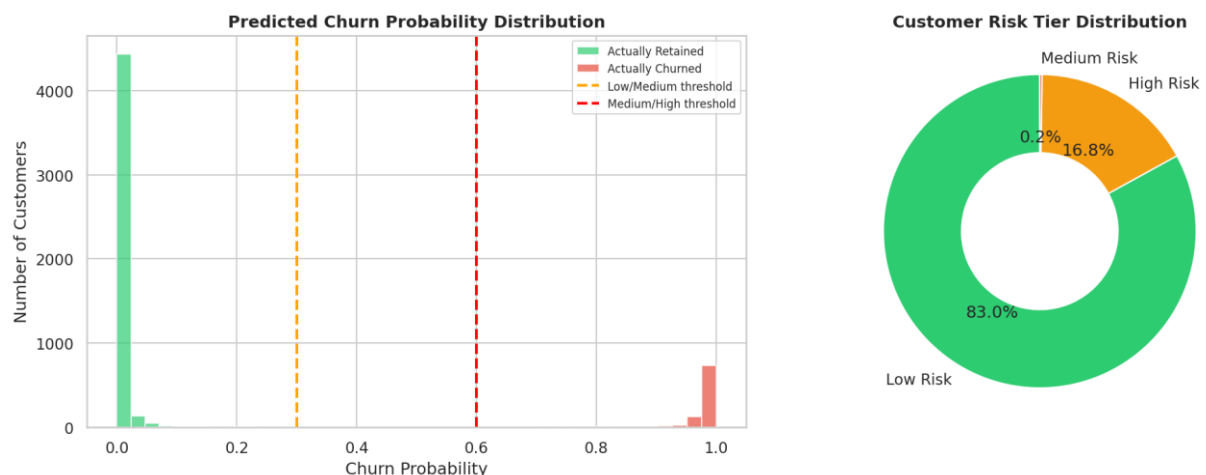


Fig 4 — Churn Probability Distribution & Customer Risk Tier Breakdown

Risk Tier	Threshold	Customer Count	% of Base
Low Risk	Score < 0.30	4,672	83.0%
Medium Risk	0.30 – 0.60	13	0.2%
High Risk	Score > 0.60	945	16.8%

The 945 High Risk customers represent the immediate intervention priority. These customers are scoring above 0.60 churn probability and include the majority of the 948 confirmed churners plus at-risk currently-retained customers.

4. Top 15 Churn Drivers (XGBoost Feature Importance)

The XGBoost model ranks 15 features by their contribution to churn prediction accuracy. Tenure accounts for 20.09% of total predictive power, making it by far the most important single signal. The top four features collectively explain ~43% of model predictions.

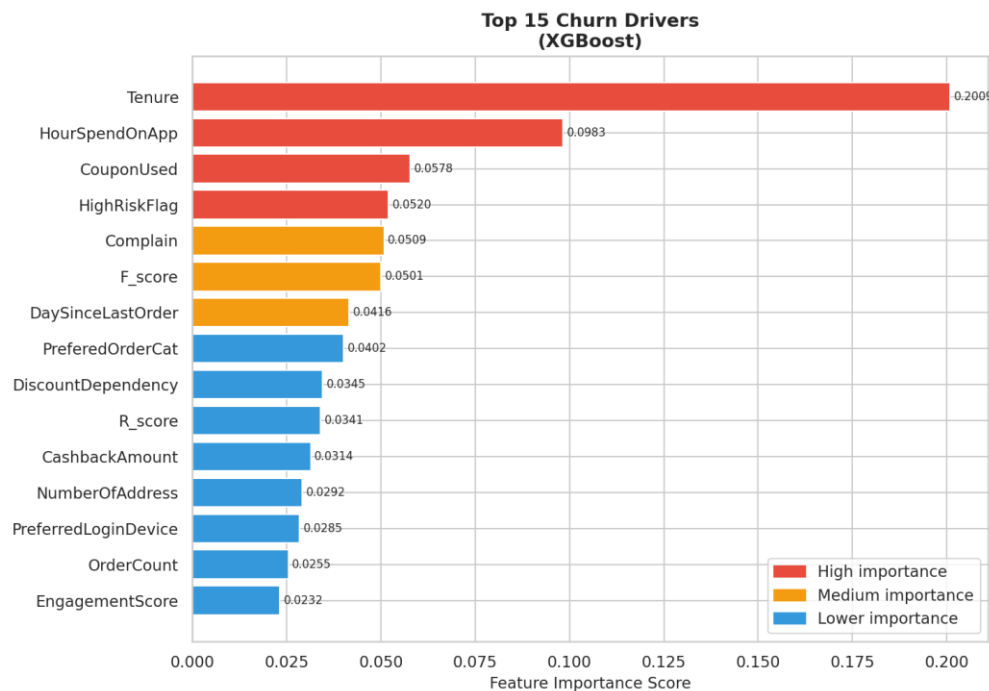


Fig 5 — Top 15 Churn Drivers — XGBoost Feature Importance

4.1 Top 10 Features Ranked

Rank	Feature	Importance	Business Interpretation
1	Tenure	0.2009	New customers churn at 51.8%; loyal (2+yr) at 0%
2	HourSpendOnApp	0.0983	Low app engagement = high churn risk
3	CouponUsed	0.0578	Discount-dependent customers leave when promos end
4	HighRiskFlag	0.0520	Compound engineered signal: complaint+low sat+recency
5	Complain	0.0509	Complainers churn at 31.67% vs 10.93%
6	F_score (Frequency)	0.0501	High order frequency builds platform habit
7	DaySinceLastOrder	0.0416	Extended inactivity = competitor migration signal
8	PreferredOrderCat	0.0402	Mobile Phone buyers: 27.4% churn vs Grocery: 4.88%
9	DiscountDependency	0.0345	Top-quartile coupon users — fragile loyalty
10	R_score (Recency)	0.0341	Recency decline precedes churn

5. Power BI Dashboard Suite

Six interactive Power BI dashboards translate analytical outputs into business-ready visual intelligence. Each dashboard includes slicers, KPI cards, and cross-filtering capability.

Dashboard 1 — Executive Dashboard

Provides a top-level view for leadership: overall churn rate, churned/retained customer counts, churn by complaint status, city tier, payment mode, and order category.

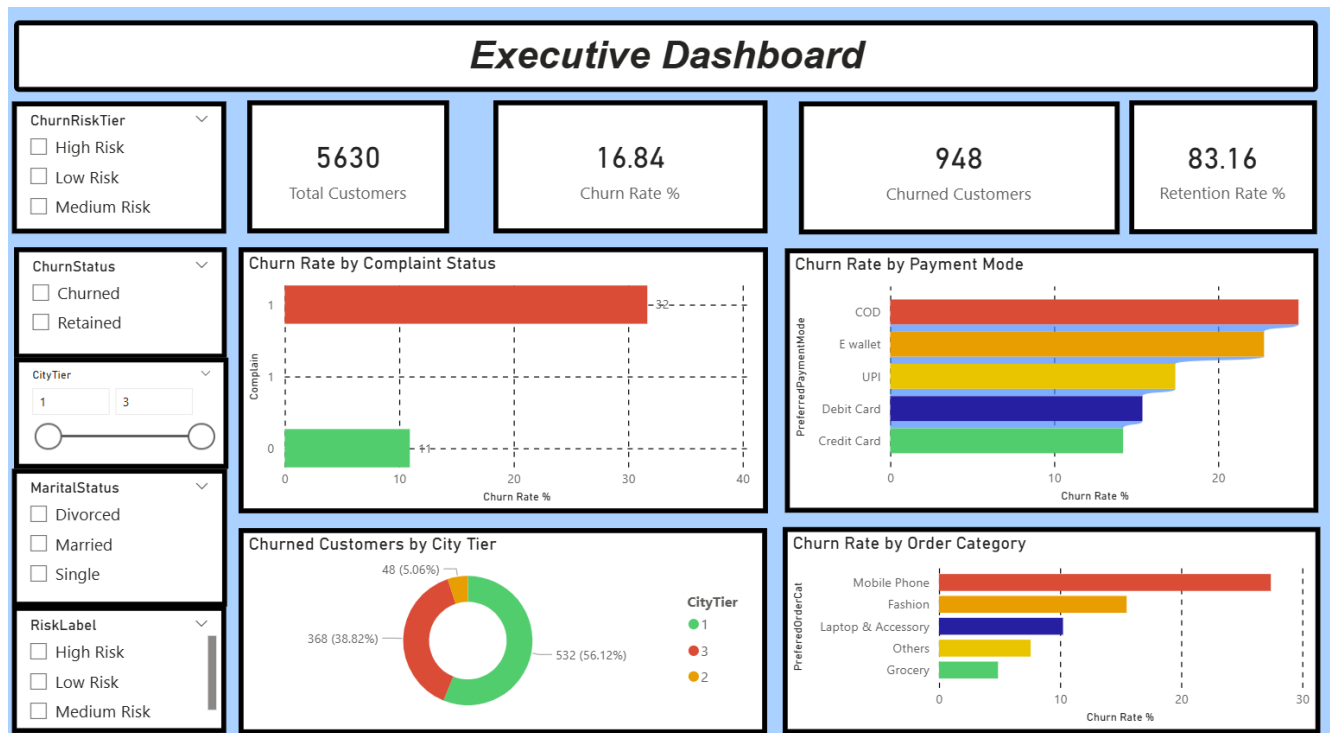


Fig 6 — Executive Dashboard (Power BI)

KPI	Value
Total Customers (Power BI view)	5,630
Churn Rate %	16.84%
Churned Customers	948
Retention Rate %	83.16%

Dashboard 2 — Retention Funnel Dashboard

Shows the customer retention funnel across tenure bands, days-since-last-order distribution, churn rate across the customer lifecycle, and churn by login device.

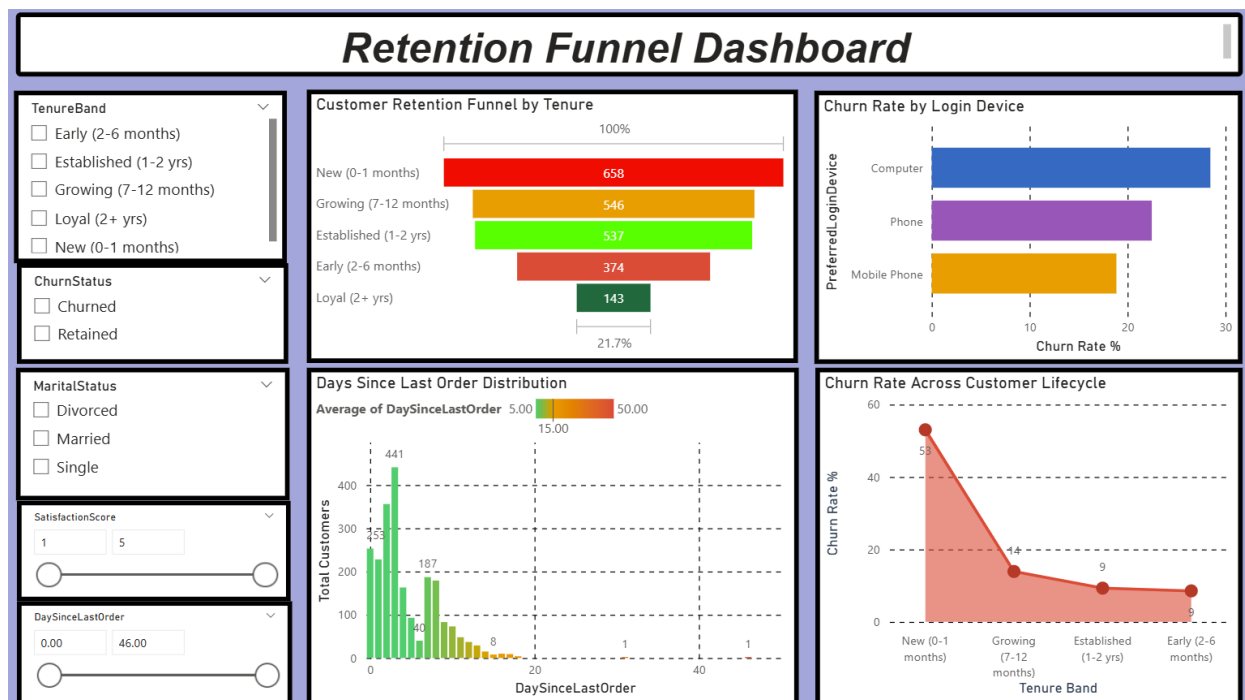


Fig 7 — Retention Funnel Dashboard (Power BI)

Tenure Band	Customers	Churn Rate
New (0–1 months)	1,198	51.84%
Early (2–6 months)	952	7.98%
Growing (7–12 months)	1,584	9.85%
Established (1–2 years)	1,467	6.48%
Loyal (2+ years)	429	0.00%

Dashboard 3 — CLV Analysis Dashboard

Analyses Customer Lifetime Value segments, revenue distribution, churn risk by CLV band, and identifies top high-value churned customers.

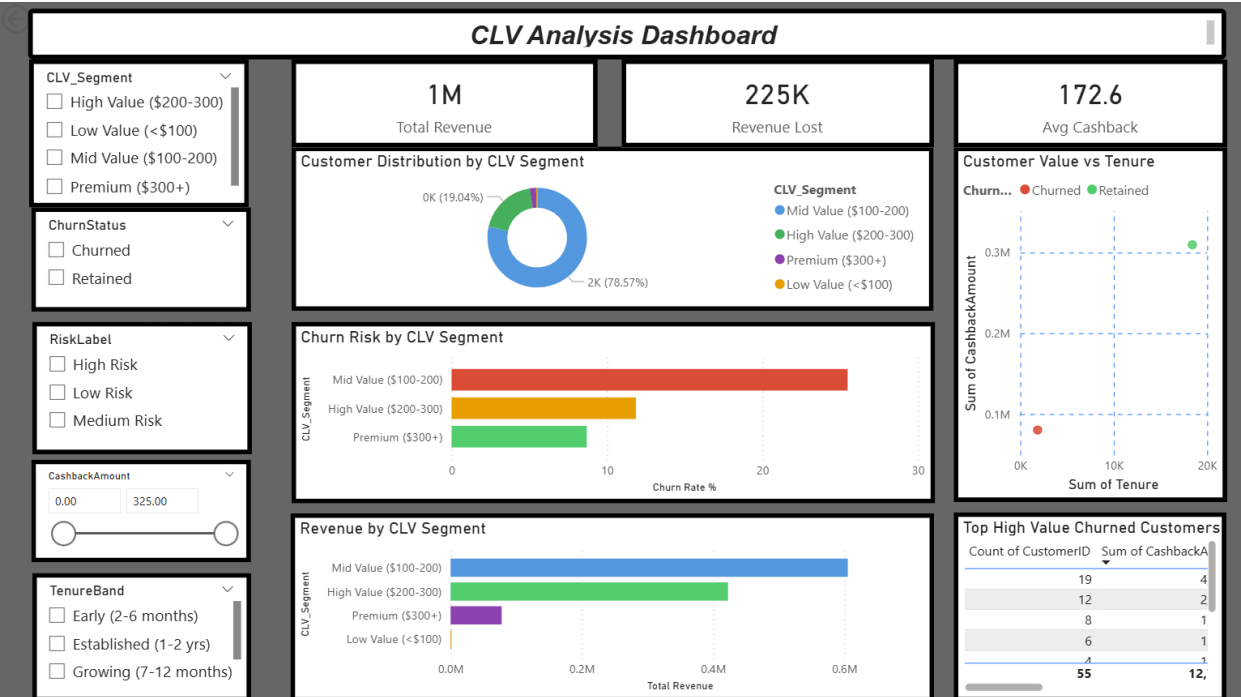


Fig 8 — CLV Analysis Dashboard (Power BI)

CLV Segment	Cashback Range	Customer Count	Churn Rate
Low Value	<\$100	12	0.00%
Mid Value	\$100–\$200	4,295	18.86%
High Value	\$200–\$300	1,167	10.97%
Premium	\$300+	156	6.41%

Total revenue (cashback): INR 997,766. Revenue lost to churned customers: INR 152,032. Revenue at risk from High Risk customers: INR 151,138.

Dashboard 4 — Repeat Purchase Dashboard

Examines purchase frequency distribution by category, churn by frequency tier, repeat vs one-time customer split, and average coupon usage by frequency group.

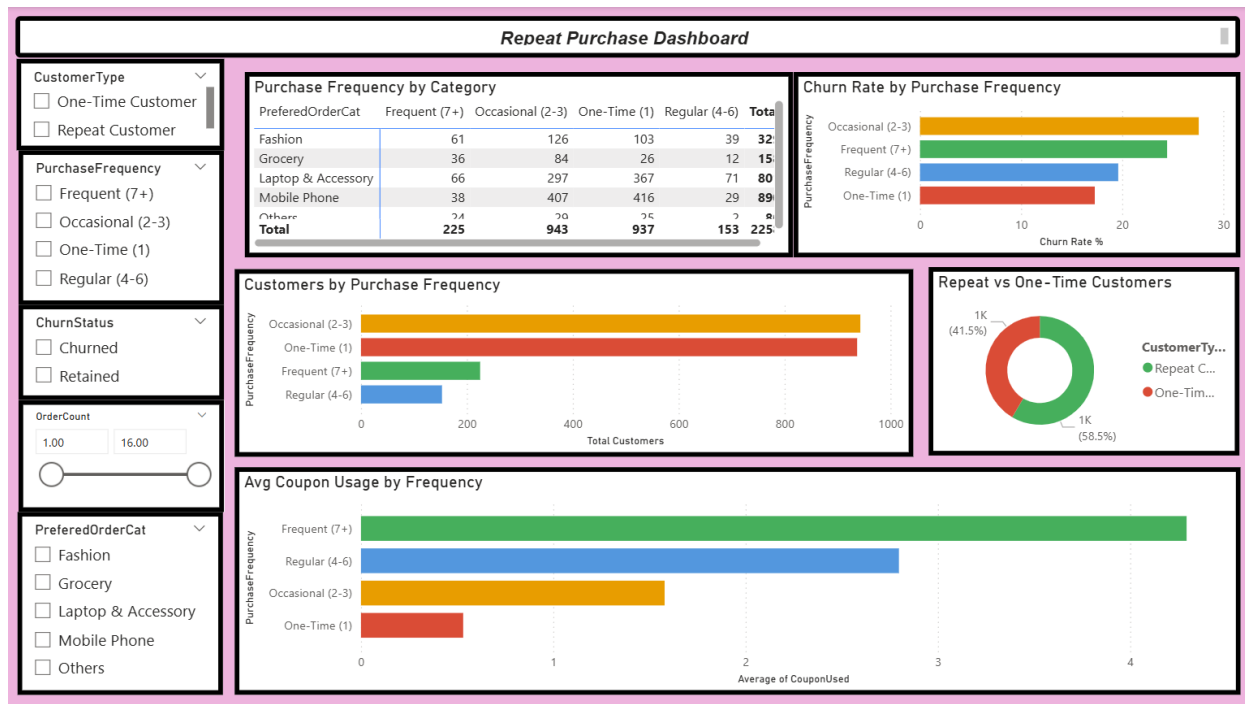


Fig 9 — Repeat Purchase Dashboard (Power BI)

Purchase Frequency	Customers	Churn Rate	Avg Coupons Used
One-Time (1 order)	937	~18%	Low
Occasional (2–3)	943	~28%	Moderate
Regular (4–6)	153	~11%	High
Frequent (7+)	225	~8%	Highest

Repeat customers: 3,879 (68.9%). One-time customers: 1,751 (31.1%). Frequent buyers (7+ orders) exhibit the lowest churn and highest coupon usage, confirming that promotional engagement builds platform habit rather than undermining loyalty at high frequency.

Dashboard 5 — RFM Segmentation Dashboard

Segments customers into five RFM groups — Champions, Loyal, At Risk, Hibernating, Lost — displaying churn rate, cashback value, frequency vs monetary scatter, and segment customer counts.

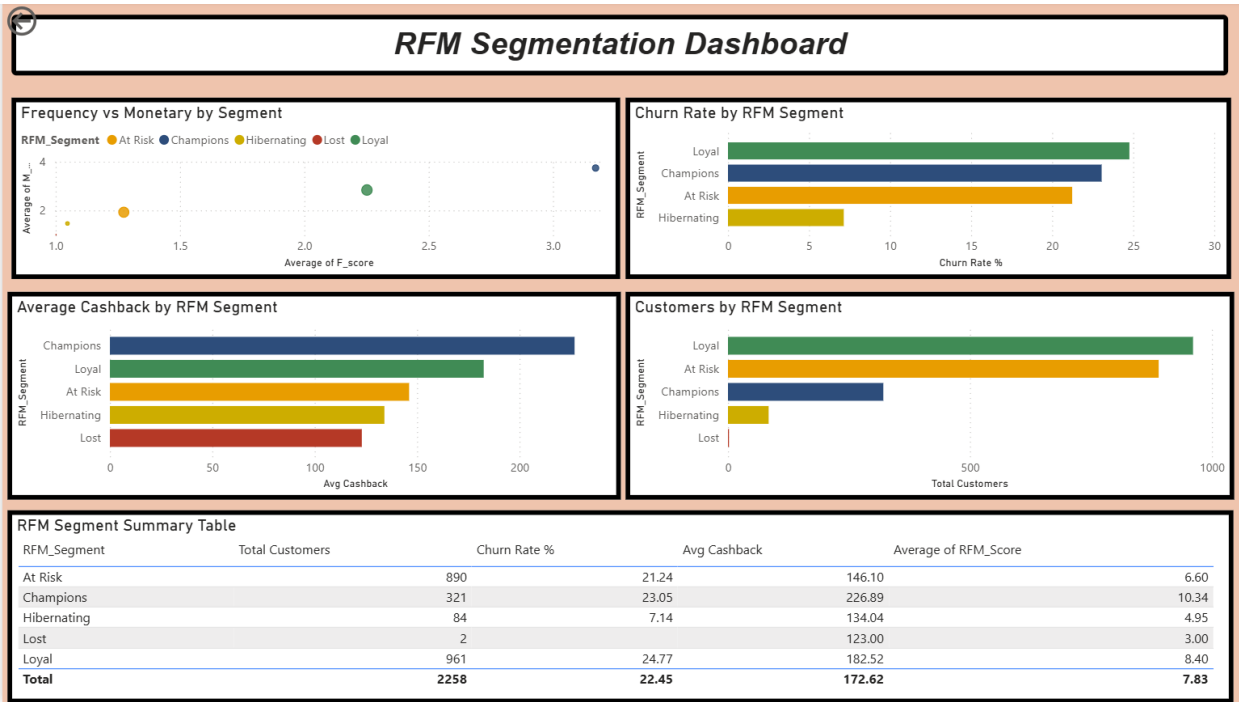


Fig 10 — RFM Segmentation Dashboard (Power BI)

RFM Segment	Customers	Churn Rate	Avg Cashback	Avg RFM Score
At Risk	2,129	19.30%	INR 154.22	6.60
Champions	773	14.10%	INR 224.46	10.34
Hibernating	474	13.08%	INR 142.39	4.95
Lost	32	3.12%	INR 137.17	3.00
Loyal	2,222	16.43%	INR 190.84	8.40
Total	5,630	16.84%	INR 172.62	7.83

Dashboard 6 — Predictive Churn Monitoring Dashboard

The operational core of the suite. Displays 240 at-risk customers by churn score, risk tier distribution, revenue at risk, and actual churn rate per risk tier — enabling the CX team to identify specific individuals for outreach.

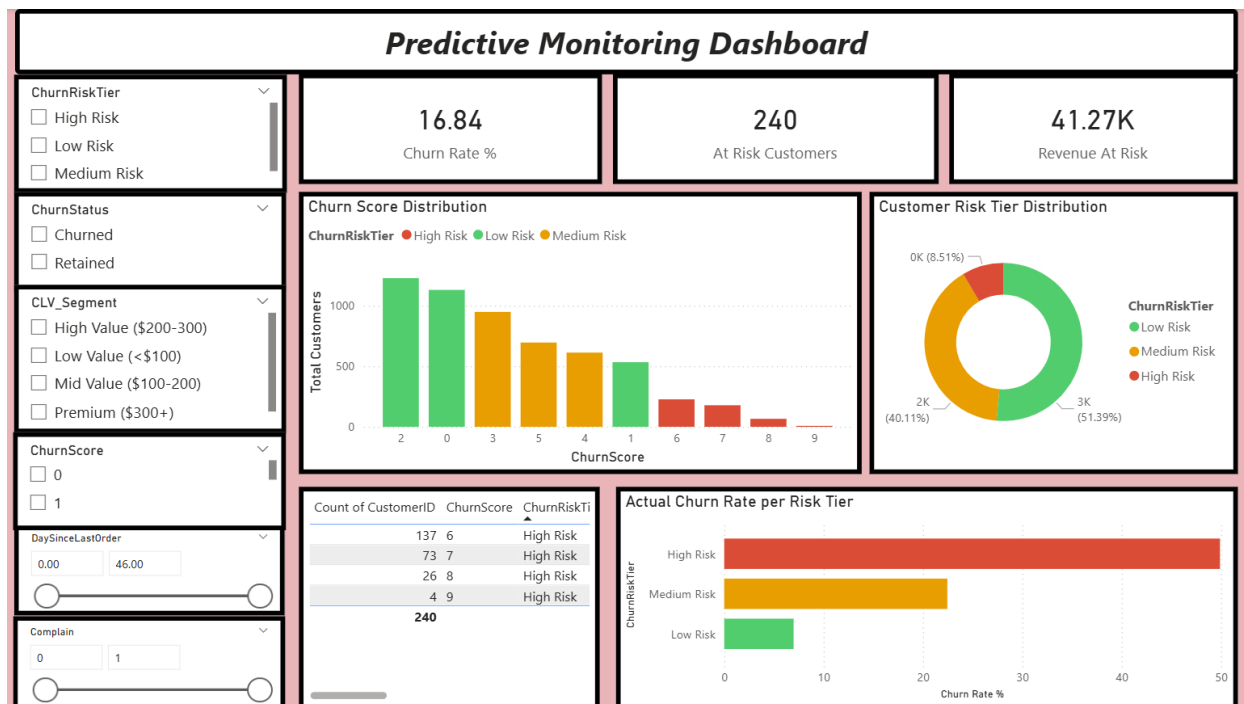


Fig 11 — Predictive Churn Monitoring Dashboard (Power BI)

Metric	Value
Churn Rate %	16.84%
At Risk Customers	240
Revenue At Risk	INR 41,270
High Risk Tier Churn Rate	~45–50%
Medium Risk Tier Churn	~20–25%
Low Risk Tier Churn	~6–7%

6. Key Findings & Business Insights

INR 152K Revenue Lost (Churned)	945 High Risk Customers	INR 151K Revenue at Risk Now	51.84% New Customer Churn
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Finding 1 — Tenure is the #1 Churn Predictor (Importance: 0.2009)

New customers (0–1 month) churn at 51.84%. Customers loyal for 2+ years churn at 0%. The 51-point gap across tenure bands confirms that churn is fundamentally a loyalty formation problem in the first 90 days, not a platform quality problem.

Finding 2 — Complaint Resolution is the Fastest Retention Lever

Complainers churn at 31.67% vs. 10.93% for satisfied customers — a 2.9× multiplier. Complaint status (importance: 0.0509) is the 5th most powerful predictor. A 24-hour complaint resolution SLA is the highest-ROI single intervention available.

Finding 3 — Mobile Phone Category Needs Urgent Attention

Mobile phone buyers churn at 27.40% — the highest of any product category, 22.5 points above Grocery (4.88%). Post-purchase dissatisfaction with high-consideration electronics purchases is the primary driver.

Finding 4 — App Engagement Prevents Churn (Importance: 0.0983)

HourSpendOnApp is the 2nd most important churn predictor. Low app engagement signals platform disengagement weeks before a customer leaves. Product features that increase session time directly reduce churn probability.

Finding 5 — Champions & Loyal Segments Churn Above Platform Average

Champions (14.10% churn, avg cashback INR 224.46) and Loyal customers (16.43% churn, avg cashback INR 190.84) are the platform's best customers — and they are churning at or above the 16.84% platform average. Differentiated VIP treatment is urgently needed.

Finding 6 — Discount Dependency Creates Fragile Retention

CouponUsed (importance: 0.0578) is the 3rd most important predictor. High coupon dependency flags customers whose loyalty is tied to promotional incentives, not platform value — making them highly vulnerable to competitor discount campaigns.

Finding 7 — Payment Mode Signals Commitment Level

Payment Mode	Churn Rate	Commitment Level
COD	24.90%	Lowest — no digital investment

E-Wallet	22.80%	Low
UPI	17.39%	Moderate
Debit Card	15.38%	Moderate
Credit Card	14.21%	Highest — most committed customers

7. Retention Strategy — Four Priority Actions

Priority 1 — Complaint Resolution SLA

Data: Complainers churn at 31.67% vs. 10.93% for non-complainers. Complain has a 0.0509 importance score.

Action: Implement 24-hour complaint acknowledgement + 72-hour resolution SLA. Assign retention specialists to High Value and Premium complainers. Automated 7-day follow-up satisfaction check.

Impact: Reducing complaint-segment churn from 31.67% to 20% would retain ~110 customers annually, recovering ~INR 16,950 in direct CLV.

Priority 2 — Mobile Phone Category Quality Programme

Data: Mobile phone buyers churn at 27.40% — highest category. PreferredOrderCat importance: 0.0402.

Action: 72-hour post-delivery quality check-in. Enhanced 30-day no-questions return window for mobile phones. Vendor quality audit on top-5 suppliers by complaint volume. Bundle 'Extended Protection Plan' with purchases.

Impact: Reducing mobile churn from 27.40% to 18% retains ~45 customers per cohort.

Priority 3 — Early-Tenure Onboarding Programme (0–90 Days)

Data: New customers (0–1 month) churn at 51.84%. Tenure is the #1 predictor (importance 0.2009). If a customer survives Month 6, churn drops to 9.85%; by Year 2, to 0%.

Action: 90-day journey — Day 1: welcome email + first-order cashback. Day 7: category recommendation. Day 21: free delivery upgrade. Day 45: loyalty tier preview. Day 60: satisfaction pulse survey (CX callback if score ≤ 3). Day 90: VIP unlock with enhanced cashback.

Impact: A 10% improvement in Month-0 retention retains ~62 additional new customers per cohort, each with multi-year CLV potential.

Priority 4 — CLV-Weighted Loyalty Programme (RFM-Based)

Data: Champions (773 customers, INR 224.46 avg cashback, 14.10% churn) and Loyal customers (2,222 customers, INR 190.84 avg cashback, 16.43% churn) are churning at platform-average rates despite being highest-value.

Action: Champions → White-glove tier: dedicated CX line, exclusive early access, maximum cashback, quarterly gift. Loyal → Gold tier: elevated cashback, priority delivery, monthly rewards. At Risk (2,129 customers) → win-back coupon campaign with time-limited bonus.

Impact: A 3-point churn reduction in the Champions segment retains 23 additional customers recovering INR 5,163 in direct CLV, with compounding long-term returns.

8. Conclusion

This study demonstrates that e-commerce customer churn is predictable, financially quantifiable, and preventable through data-driven analytics. The XGBoost model achieves 99.80% ROC-AUC, correctly identifying 181 of 190 churners in the test set with only 9 misses — providing a deployment-ready churn scoring system for the organisation.

The central finding is clear: churn is a loyalty formation failure in the first 90 days. New customers churn at 51.84%; customers who stay for 2+ years churn at 0%. Every other churn driver — complaint history, app disengagement, coupon dependency, mobile phone dissatisfaction — is an expression of this underlying reality: customers who have not yet formed a habitual relationship with the platform are one bad experience away from leaving.

The six Power BI dashboards translate these analytical findings into actionable visual intelligence for marketing, CX, finance, and executive stakeholders. The Predictive Monitoring Dashboard identifies 945 high-risk customers and INR 151,138 in revenue currently at risk — giving the retention team a precise, data-driven action list rather than a generalised campaign.

99.80% XGBoost AUC	16.84% Platform Churn Rate	945 High Risk Customers	4 Priority Retention Plans
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Tools & Technologies

Tool	Role	Key Deliverable
MySQL Workbench	SQL Analytics	8 analytical tasks — churn rates, CLV, frequency, repeat purchase, at-risk identification
Python (Google Colab)	EDA + ML Modelling	EDA dashboard, correlation heatmap, feature importance, XGBoost model, churn scores CSV
Power BI Desktop	Business Dashboards	6 interactive dashboards with slicers, KPI cards, drill-through, DAX measures